ISE 218: Fundamentals of Information Technology

Stony Brook Institute at Anhui University

Homework #2

Spring 2025

Assignment Due: March 07, 2025

Neatly write or type your answers to the following problems on a separate sheet of paper and submit a digital copy on Microsoft Teams by the due date.

For mathematical calculations you must show all work to receive full credit and simplify your answers as much as possible, giving a single number as your answer.

Please clearly print or type your **full name** and **ID #** on the first page of your submission.

For all questions involving sizes, you should use 2-based units of size. For example, 1 KB = 2^{10} bytes, not 10^3 bytes.

1. [2 pts] Convert the binary number 1101101_2 to decimal.

   $1101101_2 = 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
   $= 64 + 32 + 0 + 8 + 4 + 0 + 1$
   $= 109_{10}$

2. [2 pts] Convert the hexadecimal number $4F2A_{16}$ to binary.

   Each hexadecimal digit is converted to a 4 - bit binary number.
   $4_{16} = 0100_2$
   $F_{16} = 1111_2$
   $2_{16} = 0010_2$
   $A_{16} = 1010_2$
   $4F2A_{16} = 0100111100101010_2$

3. [2 pts] Convert the binary number 1011010010011_2 to hexadecimal.

   Group the binary digits into sets of 4 from right to left:

1011010010011_2 = 0001 0110 1001 0011_2 (add leading zeros to complete the groups)
Now convert each group to hexadecimal:
$0001_2 = 1_{16}$
$0110_2 = 6_{16}$
$1001_2 = 9_{16}$
$0011_2 = 3_{16}$
Therefore, $1011010010011_2 = 1693_{16}$

4. [2 pts] Convert the binary number 110010.101_2 to decimal.

$110010.101_2 = 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$
$= 32 + 16 + 0 + 0 + 2 + 0 + 0.5 + 0 + 0.125$
$= 50.625_{10}$

5. [2 pts] Convert the decimal number 26_{10} to hexadecimal.

$26_{10} = 1 \times 16^1 + 10 \times 16^0$
$= 1 \times 16 + 10 \times 1$
$= 1A_{16}$

6. [2 pts] Convert the hexadecimal number $4FA_{16}$ to decimal.

$4FA_{16} = 4 \times 16^2 + 15 \times 16^1 + 10 \times 16^0$
$= 4 \times 256 + 15 \times 16 + 10 \times 1$
$= 1024 + 240 + 10$
$= 1274_{10}$

7. [2 pts] Perform the following addition of 8-bit two's complement numbers: 11010110 + 11011111

Solution1:
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  11010110
+ 11011111
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 110110101
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Since this is an 8-bit addition, we discard the overflow bit:

Result = 10110101_2

Solution2:
First, convert to decimal:

$11010110_2 = -42_{10}$, 110111112 = -33_{10}

Adding:

$-42 + (-33) = -75$

Convert -75 back to 8-bit two's complement:

$-75_{10} = 10110101_2$

8. [2 pts] What is the decimal value of the two's complement number 10001011?

First, check the sign bit (left-most bit). Since it is 1, it is a negative number.
Take the complement of 10001011, which is 01110100
Add 1: 01110100+1 = 01110101
$01110101_2 = 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$
$= 64 + 32 + 16 + 0 + 4 + 0 + 1$
$= 117_{10}$
Since the original number was negative, the value is -117_{10}

9. What decimal value does the 8-bit binary number 10000010 have if:

   a. [2 pts] it is interpreted as an unsigned number?

   $10000010_2 = 1 \times 2^7 + 0 \times 2^6 + 0 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$
   $= 128 + 0 + 0 + 0 + 0 + 0 + 2 + 0$
   $= 130_{10}$

   b. [2 pts] it is on a computer using signed-magnitude representation?

   The sign bit is 1, indicating a negative number.
   The magnitude is given by the remaining bits:
   $0000010_2 = 2_{10}$
   Therefore, the value is -2_{10}

   c. [2 pts] it is on a computer using two's complement representation?

   Invert the bits: 01111101
   Add 1: 01111110

Convert to decimal:

$01111110_2 = 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0$

$= 64 + 32 + 16 + 8 + 4 + 2 + 0$

$= 126_{10}$

Since the original number was negative, the value is -126_{10}

10. Represent the decimal number −48 in binary using:

a. [2 pts] 8-bit signed magnitude

First, convert 48 to binary:

$$48_{10} = 110000_2$$

Pad with leading zeros to make it 8 bits:

$$00110000_2$$

Since it is negative, set the most significant bit to 1:

$$10110000_2$$

b. [2 pts] 8-bit two's complement

First, convert 48 to binary:

$$48_{10} = 110000_2$$

Pad with leading zeros to make it 8 bits:

$$00110000_2$$

Invert the bits:

$$11001111_2$$

Add 1:

$$11010000_2$$

Therefore, -48 in 8-bit two's complement is 11010000_2

11. [2 pts] Convert the fixed-point binary value 1101011.1011_2 to floating-point notation

Normalize the binary number: $1.1010111011_2 \times 2^6$

Convert to IEEE 754 format:

Sign bit: 0 (positive)

Exponent: 6 + 127 (bias) = 133 = 10000101_2

Mantissa: 1010111011 (drop the leading 1)

Therefore, the floating-point notation is 0 10000101 $10101110110000000000000_2$

12. [3 pts] Convert the IEEE 754 32-bit representation 10000000100100000000000000000000 into decimal.

1. Sign bit: 1 (negative)
2. Exponent: $00000001_2 = 1_{10}$

   Bias = 127, so actual exponent = 1 - 127 = -126
3. Mantissa: 1.001 (normalized)

   $$\text{Value} = -1.0012 \times 2^{-126}$$
4. In decimal: $-(1 + 1/8) \times 2^{-126} = -89 \times 2^{-126} \approx -1.125 \times 2^{-126}$

13. [3 pts] Convert the number -215.25_{10} into IEEE 754 32-bit notation.

1. Convert the absolute value to binary: $215.25_{10} = 11010111.01_2$
2. Normalize the binary number: $1.101011101_2 \times 2^7$
3. Convert to IEEE 754 format:
   - Sign bit: 1 (negative)
   - Exponent: 7 + 127 (bias) = 134 = 10000110_2
   - Mantissa: 101011101 (drop the leading 1)

Therefore, the IEEE 754 32-bit notation is 1 10000110 $10101110100000000000000_2$